

Research article

Vision-based seam tracking for GMAW fillet welding based on keypoint detection deep learning model

Mobina Mobaraki ^{a,*}, Soodeh Ahani ^a, Ringo Gonzalez ^d, Kwang Moo Yi ^b, Klaske Van Heusden ^c, Guy A. Dumont ^a

^a Electrical and Computer Engineering, University of British Columbia, Vancouver campus, 2329 West Mall, Vancouver, V6T 1Z4, British Columbia, Canada

^b Computer Science, University of British Columbia, Vancouver campus, 2329 West Mall, Vancouver, V6T 1Z4, British Columbia, Canada

^c School of Engineering, University of British Columbia, Kelowna campus, 3333 University Way, Kelowna, V1V 1V7, British Columbia, Canada

^d Novarc Technologies Inc., 1225 E Keith Rd, North Vancouver, V7J 1J3, British Columbia, Canada

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ABSTRACT

Pre-programmed welding robots significantly improved the efficiency and quality of the welds in large batch production. In small and medium batch production, the robots need appropriate sensors to perform well and adapt to the changes and uncertainties in a noisy welding environment. Vision-based sensors enabled by machine learning are making it possible to sense in process previously not measurable. One challenge is developing artificial intelligent models capable of real-time seam tracking, particularly in fillet joints where visual analysis is hindered by non-perpendicular camera angles and arc reflections. In this paper, we propose a vision system that enables automated seam tracking with a collaborative robot. The vision-based deep learning classification model detects the tacks, where the seam is not visible. It is based on a keypoint detection deep learning model that addresses the challenges in distorted and noisy images of fillet joints between the pipes and flanges during the real-time Gas Metal Arc Welding to track the location of the seam in non tack images. The system is optimized for real time seam tracking by proposing the appropriate input image size. Multiple images and multiple points are also considered to provide a controllable signal of the location of the seam with less errors and outliers. Our proposed model can track the seam with more than 80 percent accuracy for errors less than 0.3 mm in fillet joints. The high accuracy of the proposed method would result in fewer flaws and defects and reduced rework, resulting in significant cost saving in manufacturing. The real-time monitoring also enables the adaptability to slight variations in gap.

1. Introduction

Inaccurate positioning of the welding torch with respect to the seam during the welding process can deteriorate the mechanical properties and overall quality of the welded structure. Low-quality welds in pipelines can be a contributing factor to the progression of cracks, part failure, and consequently catastrophic phenomena.

Monitoring the position of the seam is a vital step during the welding process to minimize these risks and ensure precise alignment between the welding torch and the seam. Accurately extracting the seam location from welding data poses a significant challenge due to the presence of highly varying parameters in welding and substantial noise interference, particularly when the welding current is high [1].

The objective of adequate torch positioning requires seam-tracking and distance control. The former is monitoring the lateral position of the torch with respect to the workpiece and adjusting the torch's

position accordingly. This deviation may be due to heat distortion, unmachined workpieces, imperfect clamp positioning, angles of the welded joints, large tolerances, and variability in gap size resulting in an asymmetric, unstable molten metal. Monitoring this gap is required for automation and preventing welding defects such as undercut which happens due to the lack of closure, porosity due to the spatial change of the molten metal, or lack of penetration due to the problems of joint alignments [2]. The latter is monitoring the vertical position of the torch with respect to the gap which is needed for automation.

To control large-batches with little variation, preprogrammed industrial robots have shown to be effective [3]. However, in small and medium batch enterprises, it may not be time and cost efficient to adapt the complex fully automated robot to various productions and real-time environmental changes, specifically in the pipe welding industry which requires high level of precision [4]. Sensors are indispensable parts of a semi-automated solutions for small and medium batches

* Corresponding author.

E-mail address: mobinamb@student.ubc.ca (M. Mobaraki).

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as they improve the adaptability by providing feedback from the uncertain welding environment. This feedback is required for automatic control [5].

Arc and vision-based sensors are the primary seam tracking sensors in the welding industry [5]. Arc sensors can determine seam's deviations by measuring the changes in the amplitude of the current and voltage signal by using Through Arc Seam Tracking (TAST) method [6]. However, this method highly depends on the joint geometry, welding position and arc sensor algorithm [7]. Also, large amounts of noise in weld current signal can deteriorate the accuracy of the measurements by this method [6].

Vision-based sensors including infrared cameras and digital cameras are the most common and accurate sensors in welding industry [8]. Previous studies mostly used these sensors to monitor the location of the seam in butt joints, where the workpieces are parallel and the camera is positioned perpendicular to the weld. However, a thermal camera would not be an efficient sensor in industry due to its high cost and the dependency of the measurements on the surface features of the workpiece [9]. Zou et al. [10] measured the deviations of the seam by monitoring the deviations between the extremum points of the molten metal in digital images. Other studies [11] tracked the position of the seam by tracking the movement of the gray centroid in RGB images. Digital cameras are divided into active and passive methods. Since active methods require an external light source such as laser, passive methods would be a more economic sensor for seam tracking in GMAW in fillet joints [12].

In pipe welding, tracking the location of the seam in fillet joints is more challenging than butt joints. The main reasons are the non-perpendicular angle of the camera with respect to the weld, the asymmetric shape of the molten metal, and the continuous reflection of the arc on the workpiece [13].

Emergence of Artificial Intelligence (AI) could tackle the robustness issues in vision-based localization [14]. Previous AI-based seam tracking models detected the location of the seam by determining the feature points of the joint. For example, Zou et al. [15] applied an unconventional dual asymmetrical Siamese network on both the initial image without arc and the full-size noisy image. They defined the highest point in the produced confidence map as the feature point in noisy images. However, this approach may fail to detect the appropriate feature point in case of inappropriate initial positioning or translation movement during the welding process. Zhang et al. [16] proposed a convolutional Neural Network (CNN) that detects the pixels on the joints by fusing the low-level and high-level spatial features of each noisy image. However, their model resulted in low accuracy in case of different welding specification [17]. Zou et al. [18] improved the robustness and accuracy to 0.5 mm error by adding the temporal features in a Convolutional Long-Short-Term-Memory (LSTM) network. The appropriate threshold error that avoids further defects depend on the welding specifications. According to welders, this value is 0.3 mm in fillet welds which has a complex geometry and sensitive molten metal.

The promising benefits of deep learning are not shown in the current studies as they mainly focus on simple images in laser welding, where the seam information is more apparent due to low noise [19]. Limited studies [18] have recently developed deep learning models to localize the seam in more challenging welding processes such as GMAW [9] with 0.66 mm [20] and 1 mm [21] threshold error. However, they are limited to butt joints with an appropriate perpendicular view of the camera. In this paper, we propose automated seam tracking in fillet joints by using a vision-based keypoint detection deep learning model to track the seam and control the distance of fillet joints between the pipes and flanges during the online GMAW process with a threshold error of 0.3 mm. In our setup, we find the seam location independent of the wire tip. The keypoint detection model detects the location of the seam and calculates the deviation between the location of the predicted point and center of the image (the wire weaves around the center of the

torch which is the same as the center of image as the torch is fixed to the camera). It considers this deviation as the initial deviation. During the welding process, it continuously calculates the deviation between the predicted point from the keypoint detection model and the center point and feeds this into a controller.

Compared with a segmentation model, a keypoint detection model would be more appropriate for our application as it focuses on the critical points more accurately than detecting all the pixels right. Therefore, the model is not only dependent on the uncertain and asymmetric boundary of the molten metal or the distorted perspective of the joints in non-perpendicular camera view. This solves the challenges of seam tracking in fillet joints. Focusing on a few points also speeds up the data labeling process. Considering the real-time behavior of the welding process, we optimize the size of the input image to prepare a real time seam tracking model. Also, given that the predicted location of the seam will be used as an input to a controller, outliers in the predicted location of the seam would have negative consequences on the controller [22]. We tackled the issue by considering the temporal features of the images, i.e. stacking the three consecutive images as an input to the model and enforcing the model to extract deeper features by adding a second measured point to define the seam location.

As a summary, our main contribution in this paper is a seam tracking model with the following features:

- We propose a deep learning tack detection model based on ResNet 101 [23] to first filter the frames where the seam is visible and then we apply the seam tracking model only on filtered images based on Stacked hourglass model [24].
- We propose an appropriate input image size to reduce the test time in the real-time welding process while maintaining the desirable accuracy.
- We propose the second-point technique to enforce the seam tracking model to extract deeper features from the images and reduce the error.
- We add the temporal features to the seam tracking model by considering the three consecutive frames as inputs to the model. We show this can further reduce the maximum error and Root Mean Square Error (RMSE) which is sensitive to outliers.

2. Related work

Keypoint detection models were developed to localize human's joints in noisy images with different human's appearances, kinematics, imaging, and ambiguity [25]. These models are mainly divided into top-down and bottom-up approaches [26]. An example of a top-down algorithm is AlphaPose model [27] which first localizes the human and then calculates the pose. A good example of bottom-up algorithms is the OpenPose model [28] which first detects the key points by predicting the confidence map and then forms the appropriate pairs by predicting the part affinity fields. Grouping the key points into individual poses would be a challenge for real-time usage. Stacked hourglass network [24] can be an appropriate model to find the location of the seam in a real time from the challenging welding images. It has significantly improved the performance of the previous keypoint detection models by extracting more and deeper features from the images. Recent studies are developed based on this architecture such as CenterNet [29], which confirms the applicability of this architecture for real-time applications. Similar to finding human body points from the images, we aim to focus on a few critical points on the noisy and distorted welding images to detect the (x, y) location of the seam. Since the enhanced developments [30,31] are designed for complex backgrounds and multi persons, they are not relevant to our application. We expect its simpler architecture [24] to provide a better balance between accuracy and reasoning time to detect the location of the seam in the real-time welding process.



Fig. 1. A collaborative welding robot and its complementary metal oxide semiconductor camera which is used to capture the welding videos.

3. Equipment and dataset

3.1. Equipment

Fig. 1 shows the collaborative Spool Welding Robot (SWR) which was used in this paper to capture the dataset. There is a Complementary Metal Oxide Semiconductor (CMOS) camera fixed to the torch (Nov-Eye™). It takes an online video of the Gas Metal Arc Welding (GMAW) process so the operator can monitor the torch's movement through a screen. There are three servo motors on the robot which enable three degrees of freedom including seam, weave, and distance motions. The operator can manually control the horizontal and vertical position of the torch.

3.2. Data collection

Collecting a sufficient and representative image dataset is a primary step in a machine learning pipeline. To consider geometrical variations, 17 welding tests were conducted on both inner and outer flanges with pipe diameters of 6", 8", and 10". In order to consider noises in welding images, we applied reflection and illumination changes in the image dataset by grinding half of the pipes with a fine grit of 200 disk size to make a smooth and shiny surface. Also, some defects such as uneven leg lengths and undercut may happen during the welding process, where the molten metal gets a different shape from the normal situation. To consider both normal and defective data, different common defects in fillet welds such as undercut, and uneven leg lengths were intentionally created. Finally, to include the effect of parameter variation in the image dataset, we changed the welding parameters as shown in Table 1. Fig. 2 shows samples of our dataset. Changes in welding parameter can also add noise in the welding images. For example, increasing the welding current results in excessive input heat and arc light which added lightening noises to the welding images.

3.3. Preliminary data analysis

3.3.1. Feasibility analysis

Consistently detecting the seam location in fillet welding images is more challenging than in butt welding images due to the continuous

Table 1

Welding parameters range during data capturing.

Welding parameters	Values
Average current (A)	70–200
Voltage (V)	17–35
Wire Feed Speed (IPM)	100–240
Travel speed (IPM)	9.5–17.5

reflection of the arc on the workpiece and the non-perpendicular angle of the camera in fillet welding images shown in Fig. 3.

In order to confirm the feasibility of detecting the location of the seam based on these challenging images, we extracted the geometrical and Haralick texture features from 100 fillet welding images. For each image, we annotated the position of the seam and applied a confusion matrix to find the percentage of correlation between these features and the (x, y) position of the seam. Table 2 shows that the center puddle (center point of the molten metal) and the angle of its legs can be the potential features to determine the x position of the seam. This table also demonstrates the center puddle and center arc (center point of the arc), as well as the texture features of the image including the dissimilarity (a measure of distance between pairs of pixels in the region of interest), correlation (a measure of the linear dependency of gray levels of neighboring pixels), and homogeneity (Inverse Difference Moment which is very large for a smaller gray tone) can be the potential features to detect the y position of the seam.

We developed a linear regression model between the extracted features and the position of the seam. This model could determine the y position with 78% accuracy and the x position with 73% accuracy for 0.3 mm error or less. The results confirm the feasibility of using images for seam estimation.

3.3.2. Investigating the representative point on welding images as the seam point and their stability

In order to determine the representative point of the seam on welding images, some points such as the bottom point of the molten metal, the bottom point of the arc, the bottom point of the wire, and the center point of the molten metal were annotated on 100 consecutive images of a welding video. The time series behavior of the (x, y) position of these points is shown in Fig. 4.

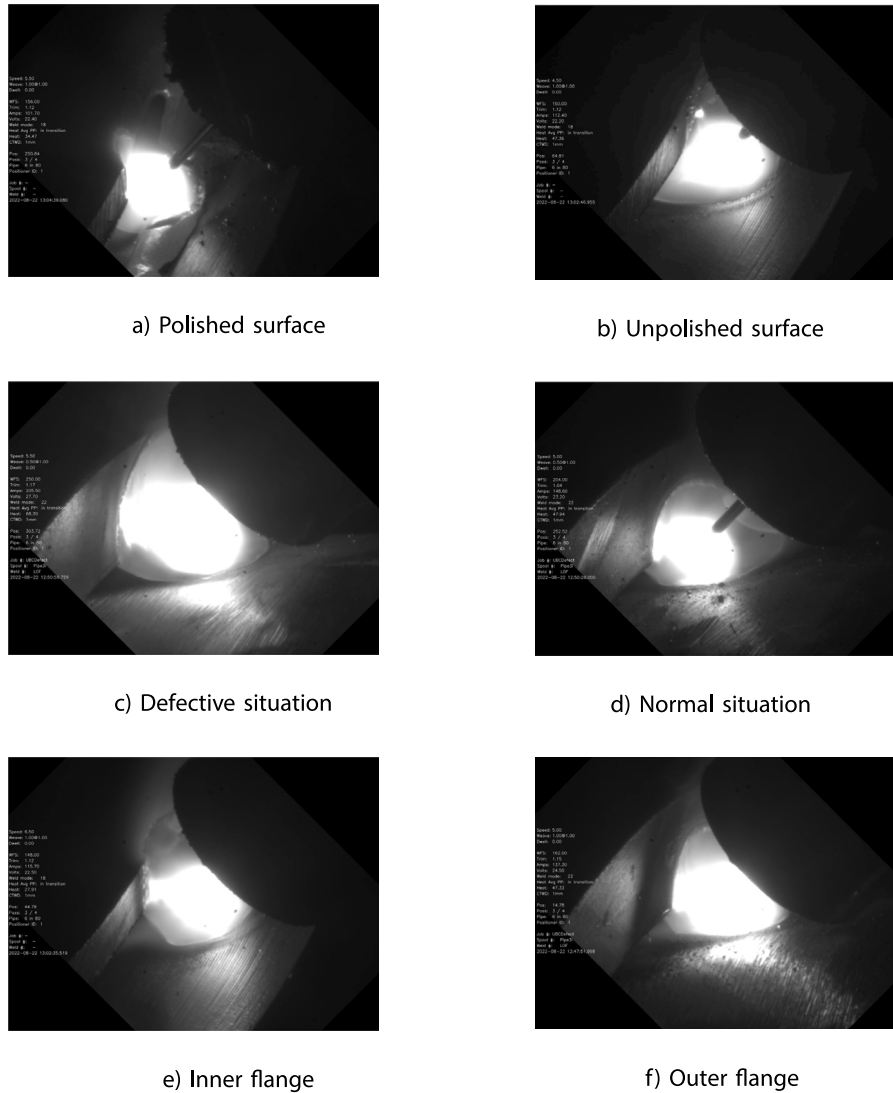


Fig. 2. Samples of the welding images that show the diversity of the dataset; including (a) polished and (b) unpolished surface, (c) defective and (d) normal situations, as well as (e) inner and (f) outer flanges.

Table 2

Percentage of the correlation between the geometrical and texture features of the welding images and the (x, y) position of the bottom point of the puddle.

Features	Correlation to bottom puddle (x)	Correlation to bottom puddle (y)
center puddle	0.92	0.96
center arc	<0.5	0.97
right leg angle	0.66	<0.5
left leg angle	−0.62	<0.5
dissimilarity	<0.5	0.86
correlation	<0.5	0.73
homogeneity	<0.5	0.88
energy	<0.5	−0.59

In order to define a controllable position of the seam, it should be a stable point with the minimum number of sudden jumps in the time-series behavior of the location of the desired point known as structural breaks. Also, in order to achieve consistent labeling, this point should be visible and easily detectable.

The center point of the molten metal in Fig. 4 would not be an consistent representative point for the seam point due to the uncertain and asymmetrical shape of the molten metal. The bottom point of the arc and wire would be easily detectable via image processing and computer vision techniques but these points are not controllable.

Therefore, the bottom point of the molten metal would be the most visible and stable point in the welding images which is controllable.

3.4. Data labeling

Based on the results from Sections 3.3.1 and 3.3.2, best representative of the seam in these challenging images is the bottom point of the molten metal. To locate this point on a large number of images, we used a video annotator interface [32] to track the location of the cursor while the video is playing.

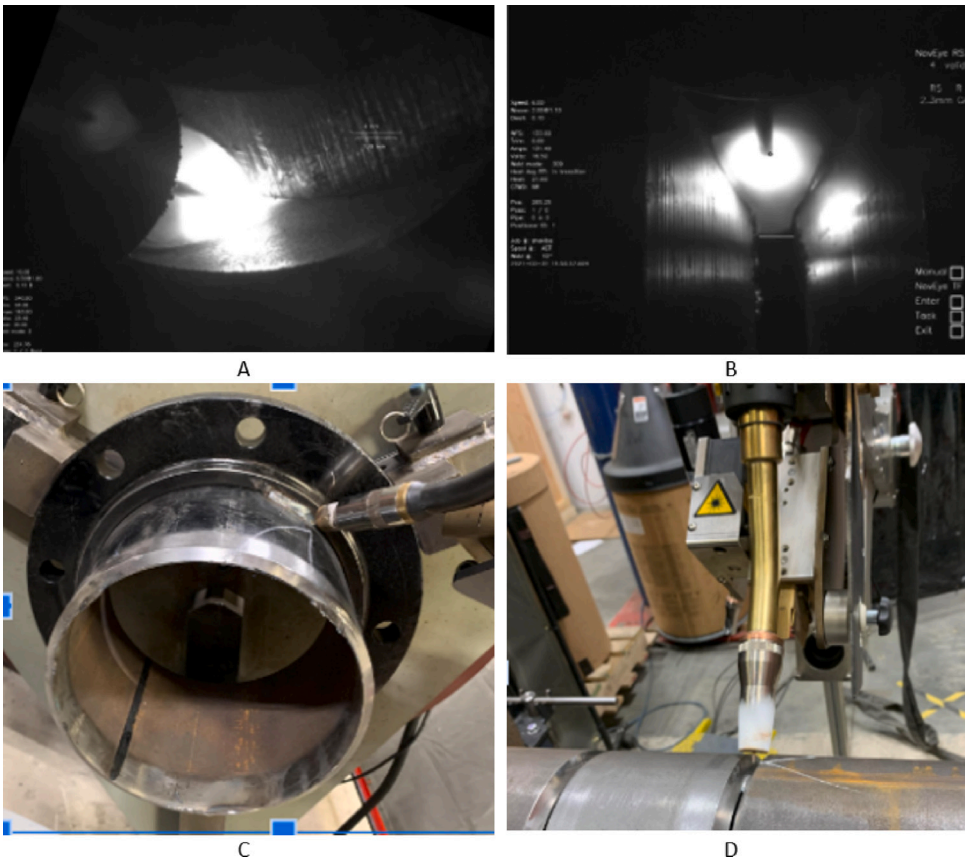


Fig. 3. The continuous reflection between the arc and the workpiece in the fillet joint (figure A) and the non-perpendicular angle of the camera in this joint (figure C) make these images more challenging to analyze with respect to the welding images in butt joints (figures B and D).

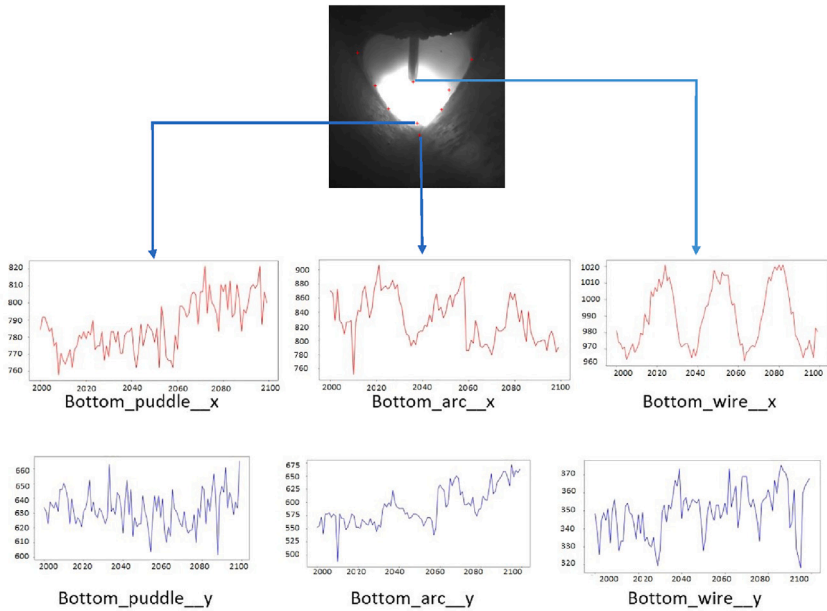


Fig. 4. Candidate points for a location of the seam on welding images and their time series behavior in 100 consecutive images of a welding video.

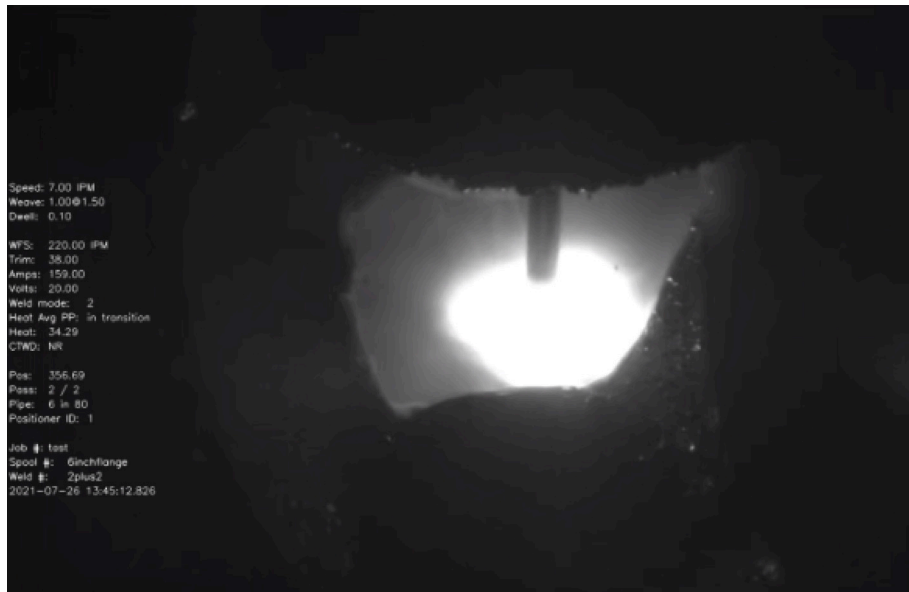
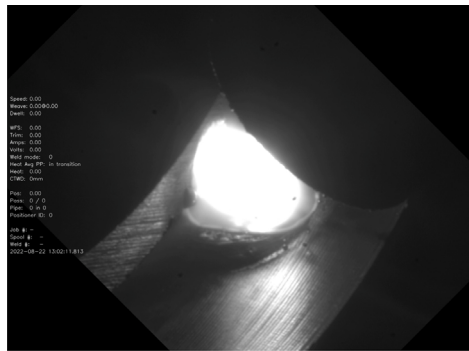
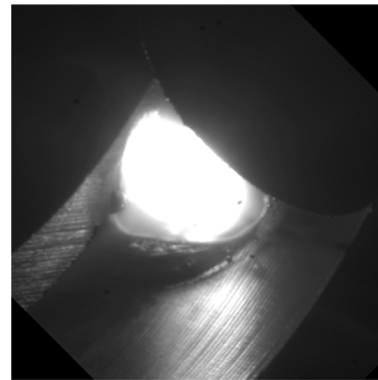


Fig. 5. A sample of video frames with tack where the seam is not visible and seam tracking should not be applied.



a) Original image



b) Cropped image

Fig. 6. Samples of the (a) original image with size of 1680×1240 and the (b) cropped image with size of 1152×1152 .

3.5. Data preprocessing

In order to prepare the labeled data as an appropriate input to the deep learning model, the following steps have been taken:

- **Cleaning the dataset:** According to our consultation with welders, seam tracking is not required during the first and last 6 s of the welding videos as the molten metal is unstable. It is also not required in the special parts of the welding video where the molten metal is on the tack and the seam line is not visible as shown in Fig. 5. We cleaned the image dataset by removing these frames from the welding videos. We used a deep learning classification model based on ResNet 101 [23] to filter the images with invisible seam.
- **Resizing:** The image size from NovEye™ was (1680×1240) . We cropped the fixed area from the original image by removing the black top area (torch) and the written text on the left side and the black area on the right of the original images and focusing on the region of interest with a size of (1152×1152) as shown in Fig. 6. We down-sampled the input image size into (256×256) as it is recommended by the Stacked Hourglass model [24] using the Inter-area interpolation method. The size of the output heatmap

is (64×64) . Fig. 7 show one sample of input image and output heatmap.

- **Normalization:** In order to reduce the effect of variations caused by illumination changes in images, we normalized the welding images to reduce the skewness. We used 0.28 and 0.25 as the mean and standard deviation to normalize our image dataset.

4. Model

In this section, the structure of the Stacked Hourglass model as well as the training information are discussed.

4.1. Model structure

Our algorithm consists of the following steps:

- First, we used a tack detection model to detect the normal images and only feed the filtered frames to the seam tracking model. We trained a classification model based on ResNet 101 and ResNet18 [23]. Since the seam is not visible on tack images, ignoring this step would result in the keypoint detection model mistakenly devotes high confidence level to the cracks on the

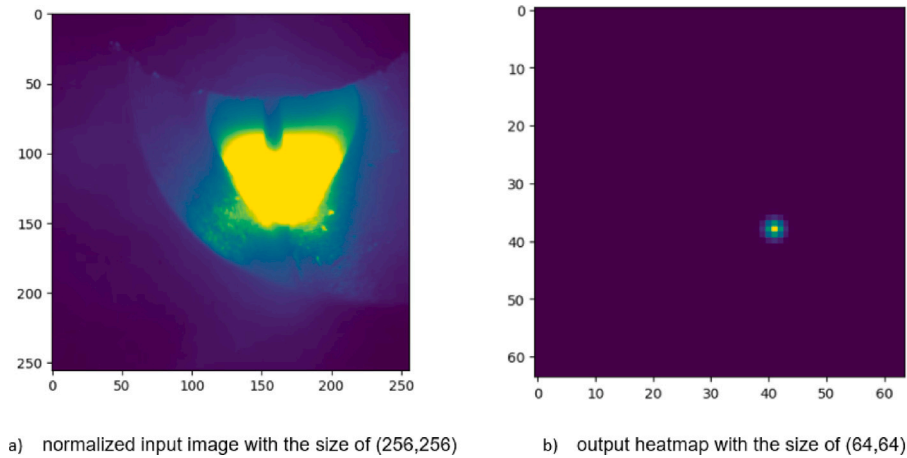


Fig. 7. Samples of (a) input and (b) output of the stacked hourglass seam tracking model.

tack and detects them as the location of the seam. Also, the tack detection model can improve the stability of the seam tracking task [33] and is used for other quality control and safety purposes during the welding process [34].

- Second, we used a human pose estimation model to find the (x, y) location of the seam on the filtered welding images. We trained a pre-trained stacked hourglass model [24] with 2 hourglasses. The pose estimation model consists of multiple hourglasses stacked to each other [24]. Each hourglass is an extensive residual module that changes the resolution of an image by passing it through a pooling and subsequent upsampling layer. This allows the model to extract deeper features from both local and global scales. The subsequent hourglass reassesses the output of the previous one, which makes the final result more accurate. Stacking multiple hourglasses produces a symmetric architecture of end-to-end bottom-up and top-down interference, which allows the model to extract both high level semantic information as well as spatial details in each image. It is also suitable for real-time applications such as welding due to the UNet structure [35,36]. Also, the output is in the form of a heatmap that shows the probability of each point at each pixel, which allows for more precise localization. To improve the performance of the seam tracking model, we focused on data augmentation [37,38] and the following techniques to extract the specific features inherent in GMAW in fillet joints. The output of this model is used as an input for a controller. To ensure an appropriate input for the controller, the number of outliers should be minimized. To address this challenge, we proposed the two following techniques:

- **Including the temporal features in the keypoint detection model:** Given the fact that the location of the seam does not drastically change in consecutive frames, adding the temporal features can reduce the possibility of outliers in the output of the keypoint detection model. Therefore, adding the features of the previous frames to the features of the current frame, can provide a more consistent prediction of the location of the seam. We concatenate three consecutive frames and make a new image with three channels with the same width and height as the original image and re-train the keypoint detection model on the concatenated images.
- **Enforcing the keypoint detection model to extract deeper features:** The predicted point should always remain on the seam curve. Considering this constraint can also reduce the possibility of outliers. We add another keypoint on the seam

curve with a constant distance to the first keypoint and re-train the keypoint detection model with these two keypoints as the labels. This enforces the model to extract more intricate features of the image relevant to the seam curve by providing additional supervision signals and consequently predict a more consistent keypoint as the location of the seam.

4.2. Training details

Tables 3 and 4 show the number of training, validation, and test datasets, the hyperparameters values, and the training time to train the tack detection and the seam tracking model, respectively.

4.3. Evaluation details

To evaluate the performance of the seam tracking model, we used Percentage of Correct Keypoint (PCK) [39] metric that measures the percentage of the predicted locations of the seam which are within a threshold distance of the ground truth. This considers the uncertainty in the ground truth labels, due to the human error in the manual labeling, by determining an interval for the ground truth point instead of a rigid (x, y) location.

5. Results and discussion

5.1. Performance results of the trained models

5.1.1. Tack detection

Fig. 8 shows the training and validation loss of the tack detection model. This model can find the tack frames with 89.62% accuracy in the test dataset. Incorrect predictions of the tack detection model are mostly due to the uncertainties in tack/normal labels. As we show, this issue can be compensated by adding temporal features to the seam tracking model.

It is necessary to use an advanced residual network to classify the tack images. Table 5 presents an ablation study to compare the ResNet 101 model with the simplified ResNet 18 model. As depicted, there is a 4% decrease in accuracy, leading to mislabeling of 2500 images. These mislabeled images would subsequently be processed by the keypoint detection model, potentially leading to wrongly detecting the cracks on the tacks as seam points, causing a significant number of outliers and wrong seam predictions. Since the reasoning time of the ResNet 101 is half of the Stacked Hourglass model, it does not significantly impact the overall reasoning time of the seam tracking system.

Table 3
Specifications for training the tack detection model.

Feature	Value	Feature	Value
Model	ResNet 101	# Unfreeze layers	1
Train time (hours)	0.5	Model volume (MB)	43.757
Learning rate	1e-5	Batch size	256
early stopping	Patience=10, epsilon=0	Loss function	MSE
# normal training images	51 335	# tack training images	9801
# normal validation images	7084	# tack validation images	558
# normal test images	5921	# tack test images	1710
GPU	NVIDIA GeForce RTX 3080		

Table 4
Specifications for training the seam tracking model.

Feature	Value	Feature	Value
Model	Stacked hourglass 2	# Unfreeze layers	all
Train time (hours)	23	Model volume (MB)	27.3
Learning rate	2.5e-5	Batch size	4
early stopping	Patience=10, epsilon=0	Loss function	MSE
Optimizer	Adam	# training images	30 906
# Validation images	10 790	# Test images	10 790
GPU	NVIDIA GeForce RTX 3080		

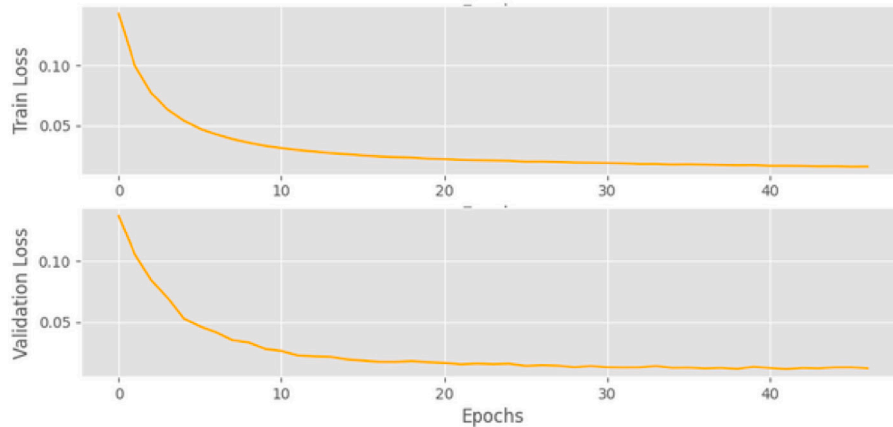


Fig. 8. Training and validation loss of the tack detection model.

Table 5

An ablation study to compare the performance of ResNet 101 and ResNet 18 models as the tack detection models.

Model	# of parameters	Reasoning time (ms)	Accuracy (%)
ResNet 101	42,504,258	1.21	89.62
ResNet 18	11,177,538	0.30	85.20

5.1.2. Seam tracking

Fig. 9 shows the training and validation loss curves. Also, Fig. 10 shows the accuracy of the seam tracking model with respect to the threshold error for x and y directions. This figure shows that we achieved more than 80% accuracy for the required threshold error of 0.3 mm in the x and y directions (the Euclidean distance between the predicted and real point).

5.2. Semi real-time implementation

To achieve the minimum test time in the real-time application, we did a semi real-time analysis on the stacked hourglass model. We reduced the input image size and evaluated the test time and model performance for each case. As Table 6 shows, for the input image size of (256×256) , the model takes 2.83 ms to go through the pixels and predict the output. Further reducing the input size to (128×128) drops the test time to 2.77 ms (2.1% drop in test time). Also, with this input size, we can achieve 80% accuracy with a threshold of less than 0.3 mm

Table 6

Comparing test time and the performance of the stacked hourglass model for three input image sizes.

Input image size	(256,256)	(128,128)	(64,64)
Test time (ms)	2.83	2.77	2.75
Threshold error for 80% accuracy in x (mm)	0.23	0.29	>1
Max error x (mm)	3.15	3.25	6.77
Threshold error for 80% accuracy in y (mm)	0.10	0.28	>1
Max error y (mm)	0.63	1.5	7.11

in both x and y directions. However, further reducing the input size to (64×64) slightly decreased the test time but had a significant negative effect on the threshold value and maximum error. With this input size, we can only achieve 80% accuracy for more than 0.3 mm threshold value and also we will get the maximum error of about 7 mm in the predictions which is 7 times larger than (128×128) . Therefore, we selected (128×128) as the optimal input size. Each pixel in this image size represents 0.207 mm. We calculated this scaling ratio by measuring the length of the wire in mm and the number of pixels that correspond to the wire in welding images.

5.3. Preparation of the results for the controller

The time-series behavior of the (x, y) location of the seam should meet the requirements for an appropriate input controller. The sheer

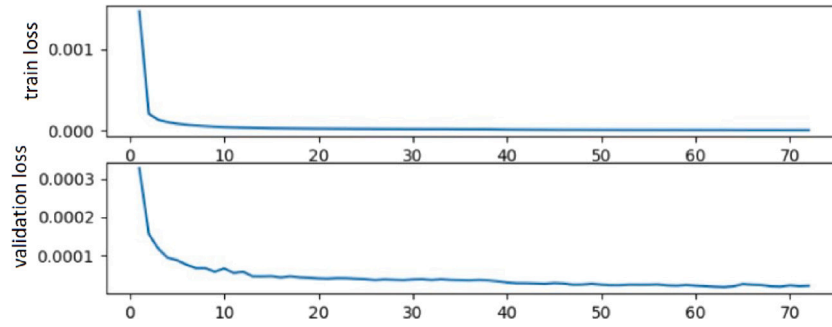


Fig. 9. Training and validation loss of the seam tracking model vs number of epochs.

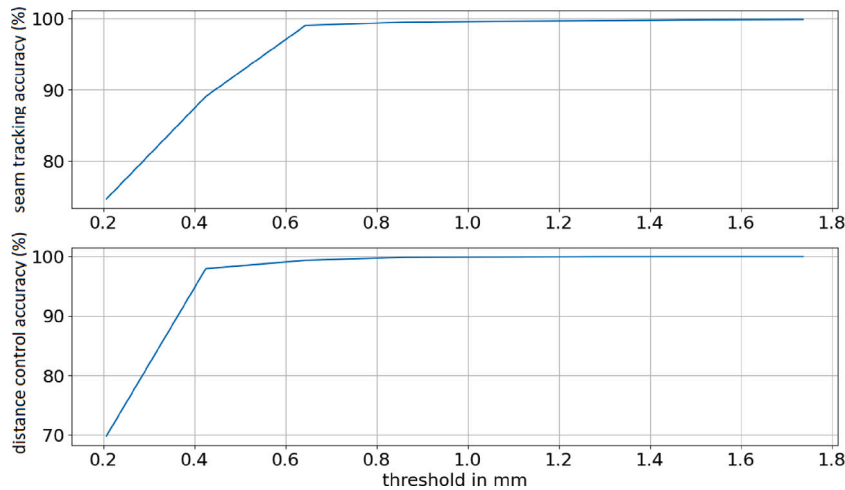


Fig. 10. Accuracy of the seam tracking model in test dataset to predict the location of the bottom point of the puddle in the x and y directions. To calculate the accuracy, the error between the prediction and ground truth is compared with different threshold error values in mm. Threshold errors less than 1 mm are acceptable.

values of the predicted point and ground truth are shown in Figs. 13 and 14. As shown in Figs. 11 and 12, the time-series of the error between the (x, y) location of the predicted point and ground truth includes outliers. The outliers in the input controller would degrade the performance of the control systems [22].

To tackle the issue, we evaluated the model's prediction in four situations on 7353 test frames:

- **Normal model:** where we applied the Stacked hourglass model on a single frame
- **Normal model with filtered situation:** where we applied a plain moving average filter with the filter size of 4 on the output of the stacked hourglass model to remove the outliers
- **Second point model:** where we added a second point on the seam curve as label and re-training the stacked hourglass model [24] on the new labeled train dataset. In this case, the stacked hourglass [24] is trained to detect two points on the images which requires deeper spatial features. By training the model to learn deeper features, we expect to reduce the number of the outliers.
- **Temporal model:** where we stacked three consecutive images and feed them as input to the stacked hourglass model [24]. Considering the location of the bottom point of the molten metal in the three images and the fourth image as label and training the stacked hourglass model [24] on the new labeled training dataset. In this case, we include the temporal features of the images so we expect to decrease the number of outliers.

We analyze the performance of the model in terms of the following criteria:

- **Threshold error to achieve 80% accuracy:** Previous studies [20,21], achieved 1 mm and 0.66 mm error threshold for butt welds. To the best of our knowledge, no study focused on seam tracking in fillet welds during the GMAW process due to the complexity of the welding process and the joint. We achieved the appropriate threshold error of 0.3 mm in fillet joints of GMAW process with about 80% accuracy. Fig. 15 compares the accuracy versus threshold error in the (x, y) location of the bottom point of the molten metal that is predicted by the stacked hourglass model in the four mentioned situations. In the x direction, adding the second point results in higher accuracy in lower threshold error values.
- **Outliers:** RMSE is one of the measures which is sensitive to outliers. Considering the relationship between the frames in the temporal model decreases the outliers in the prediction as the RMSE is reduced by 1.02% in both x and y directions.
- **Error distribution:** Figs. 16 and 17 shows the histogram of the error in the four situations. In the x direction, the maximum value of the error when the second point is added is significantly decreased from 3.25 mm to 1.01 mm (69% drop). Adding the filter and the temporal model further decreases the maximum error to 0.95 mm and 0.74 mm, respectively (77.10% and 70.77% drop). In the y direction also, the maximum error in the normal,

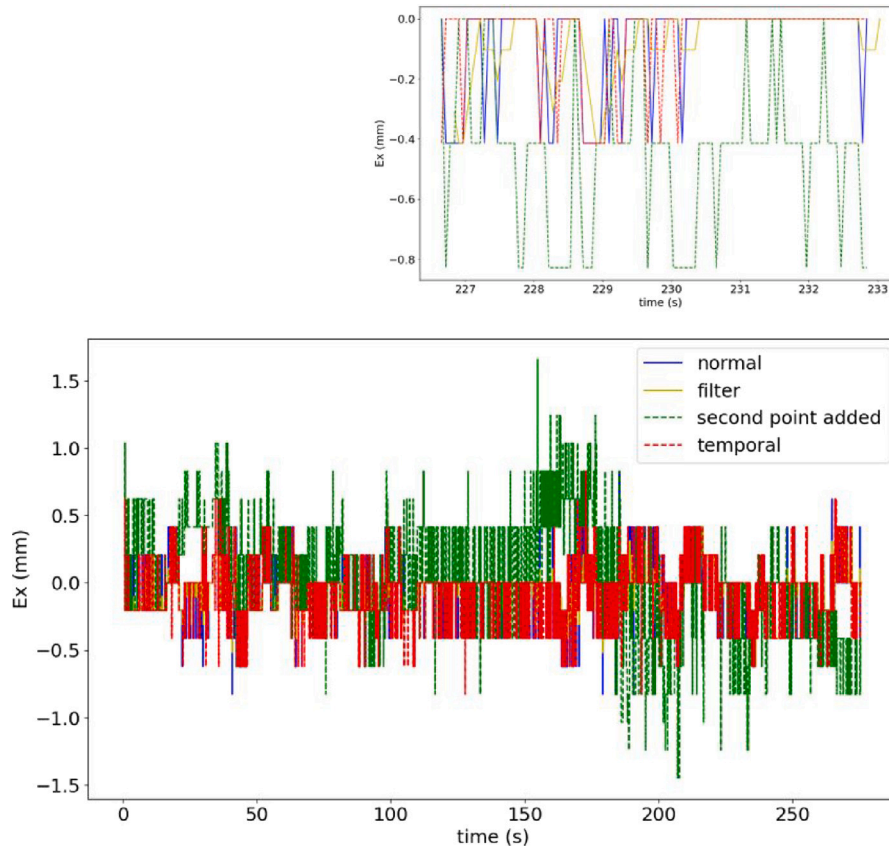


Fig. 11. Comparing the time-series behavior of the error between the ground truth x location of the bottom point of the puddle and the output prediction by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames.

second point, temporal, and filter model were 1.50 mm, 1.34, 1.11 mm, and 0.67 mm, respectively (10.67%, 26%, and 55.33% drop). Also, adding the temporal features, the second point, and filter decrease the standard deviation of the error from 0.32 to 0.31 (3.13% drop), 0.30 (6.25% drop), and 0.26 (12.5% drop), respectively in the x direction and from 0.30 to 0.32 (3.23% rise), 0.28 (12.5% drop), and 0.25 (21.88% drop), respectively in the y direction.

Table 7 summarizes our analysis.

5.4. Manufacturing benefits

Our proposed AI model advances welding manufacturing by enabling real-time seam tracking and tack detection in challenging fillet joint scenarios, where traditional models focus predominantly on post-welding analysis for butt joints. This innovation not only reduces defects, time, and costs associated with re-welding but also facilitates the use of semi-automated collaborative robots, enhancing adaptability and reducing the need for highly skilled technicians. Moreover, its real-time capabilities allow for immediate adjustments to unexpected variations in noisy welding environments, substantially improving both efficiency and quality.

6. Conclusion and future work

In conclusion, we proposed a deep learning model to track the seam and control the distance via the welding images of fillet joints

Table 7

Comparing the maximum, average, and standard deviation of the error between the (x, y) location of the bottom point of the puddle and the ground truth point that is predicted by the stacked hourglass model in three different situations including normal, adding the deeper features by including the second point on the seam as label of each image, and adding the temporal features by stacking three consecutive frames as input image and considering the location of the bottom point of the puddle in the three images as well as the upcoming fourth image.

	Normal	Filter	Second point	Temporal
Max error x (mm)	3.25	0.95	1.01	0.74
Std error x	0.32	0.26	0.30	0.31
$threshold_x$ for $> \% 80$	0.29	0.25	0.22	0.31
$RMSE_x$	0.98	0.98	0.98	0.97
Max error y (mm)	1.50	0.67	1.34	1.11
Std error y	0.30	0.25	0.28	0.32
$threshold_y$ for $> \% 80$	0.28	0.27	0.26	0.27
$RMSE_y$	0.98	0.99	0.98	0.97

between the pipes and flanges during the online GMAW process. The seam tracking and distance control model can find the seam location with more than 80 % accuracy for less than 0.3 mm error in a real-time welding process with the total reasoning time of 3.98 ms. We also showed that the temporal model performed best for seam tracking as it minimizes the maximum error to 0.74 mm. For distance control, we showed that the filter model performed best as it has the lowest maximum error of 0.67 mm, which is acceptable for most welding applications [40].

As future work, we will investigate the effect of uncertainty in ground truth labels by applying weakly-supervised learning techniques,

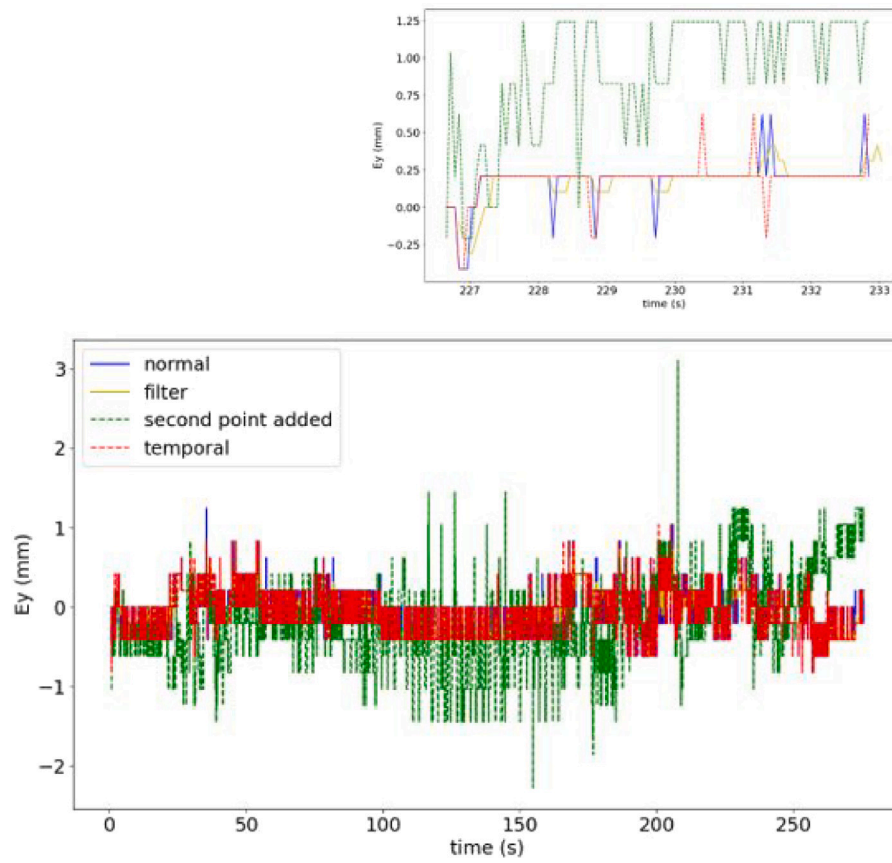


Fig. 12. Comparing the time-series behavior of the error between the ground truth y location of the bottom point of the puddle and the output prediction by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames.

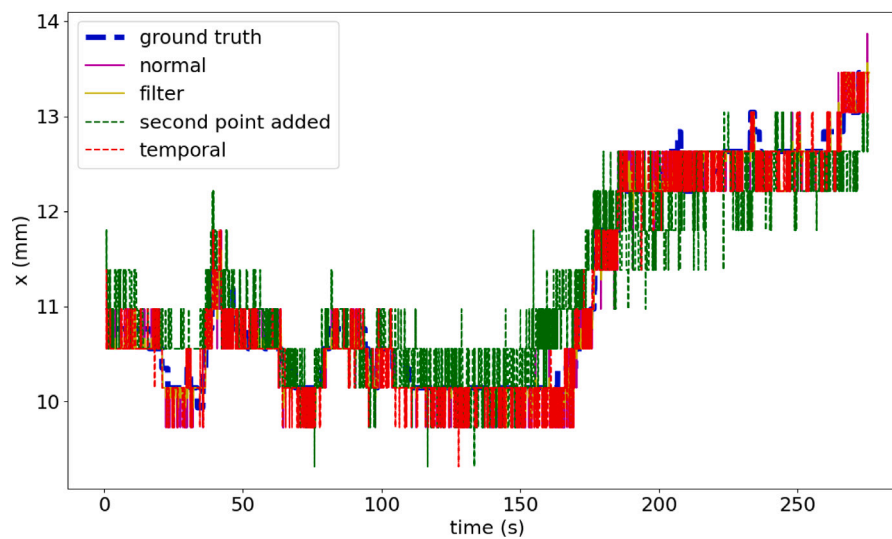


Fig. 13. Comparing the time-series behavior of the x location of the bottom point of the puddle predicted by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames.

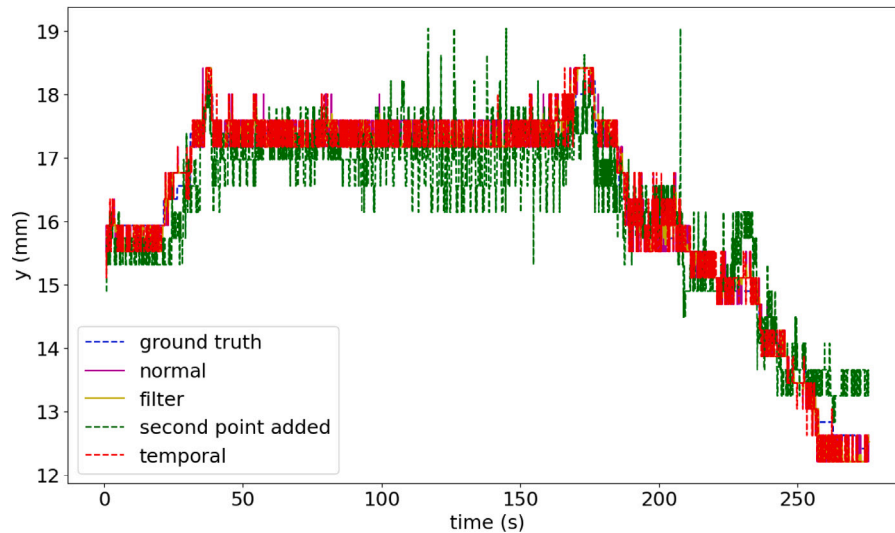


Fig. 14. Comparing the time-series behavior of the y location of the bottom point of the puddle predicted by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames.

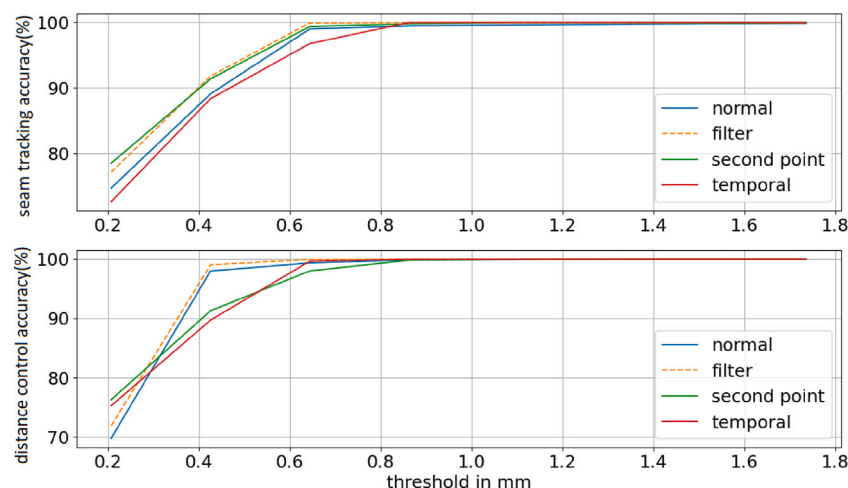


Fig. 15. Comparing the accuracy versus allowable threshold value for the error in the (x, y) location of the bottom point of the puddle predicted bottom point of the puddle by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames. Threshold errors less than 1 mm are acceptable.

in which the model learns to ignore small discrepancies between the ground truth and prediction due to an extra item in the loss function that defines a threshold for the error between the ground truth and the predictions. We will also integrate other modalities such as welding sound [41] and voltage [42] with the visual information. This leverages the performance of the seam tracking model by considering the supplementary information from other modalities in case of noise in one modality.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mobina Mobaraki reports financial support was provided by NSERC and MITACS Canada and by Novarc Technologies Inc. Guy Dumont reports a relationship with The University of British Columbia that includes: employment and reports financial support was provided by Novarc Technologies Inc. Soodeh Ahani reports financial support was

provided by The University of British Columbia and by Novarc Inc. Ringo Gonzalez reports financial support and equipment, drugs, or supplies were provided by Novarc Technologies and reports a relationship with Novarc Technologies that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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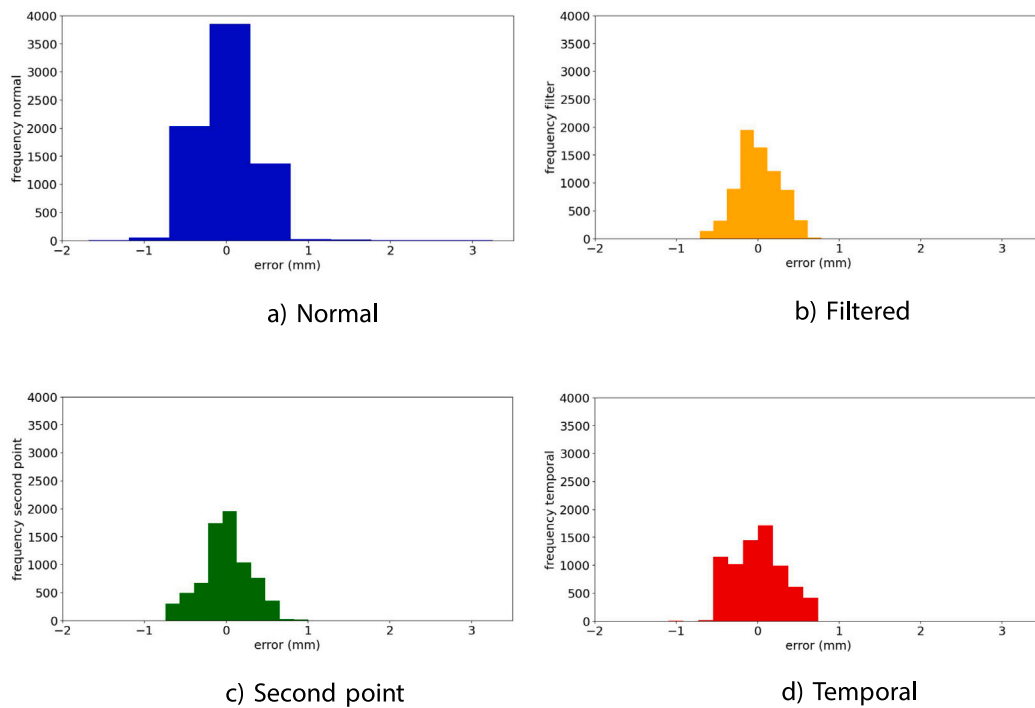


Fig. 16. Comparing the histogram of the error between the x location of the ground truth point and the predicted bottom point of the puddle by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames.

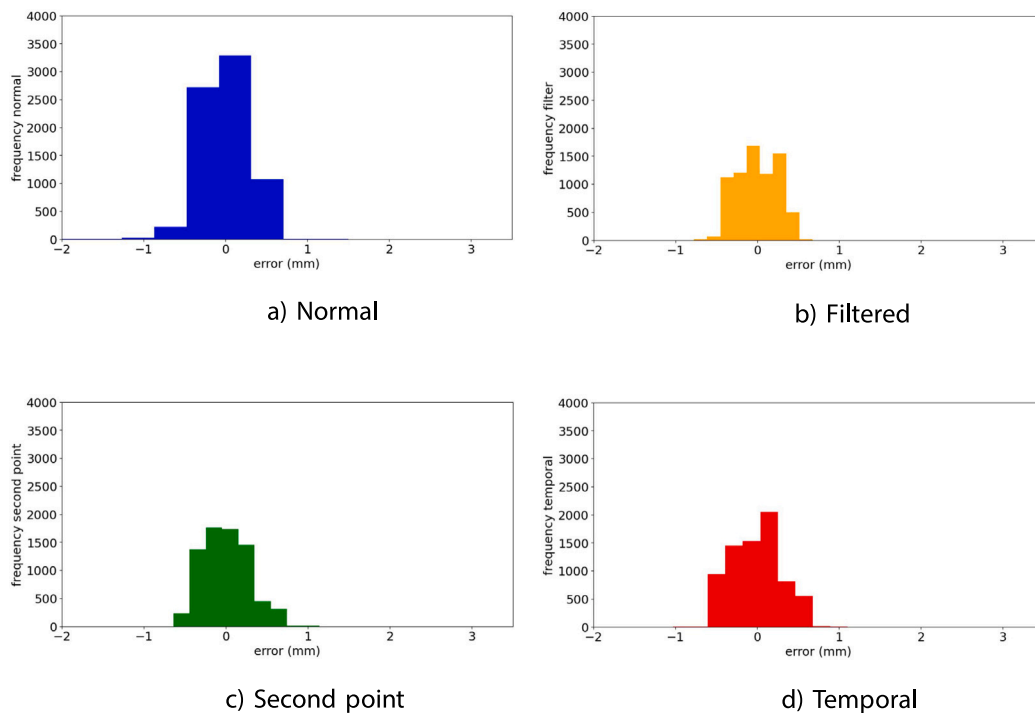


Fig. 17. Comparing the histogram of the error between the y location of the ground truth point and the predicted bottom point of the puddle by the stacked hourglass model in four different situations; including the normal model, normal model with filters, adding the second point, and considering the temporal features by stacking three consecutive frames.

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Mobina Mobaraki is currently pursuing a Ph.D. in Electrical and Computer Engineering at the University of British Columbia. She is a recipient of Mitacs Accelerate scholarship. Her current research interests include Artificial Intelligence, machine learning, and computer vision in manufacturing. She received her M.Sc. in Mechanical Engineering, applied mechanics from Sharif University of Technology in 2020 and her B.Sc. in Mechanical Engineering from Shahid Beheshti University in 2018.

Soodeh Ahani Obtaining her Ph.D. in Electrical Engineering from Sharif University of Technology in 2015, she subsequently served as a postdoctoral fellow and researcher at the University of British Columbia (UBC) from 2016. She is a data scientist, with an extensive experience in signal, image, and video processing. Her project portfolio encompasses diverse areas such as biomedical signal and video processing, defect detection, and seam tracking for robotic welding. She is also a recipient of the Mitacs Elevate, which supported a two-year partnership project with Medtronic, a global leader in the biomedical industry.

Ringo Gonzalez is the Director of Welding Innovations at Novarc, and the Welding technologist, Welding Consultant, and Welding Instructor at British Columbia Institute of Technology (BCIT). Ringo has worked internationally in diverse fields of welding technology from ship building to aerospace, military and private. He has been welding since 2001, and a Journeyman since 2006. Ringo has an “A” Level certification from Kwantlen university college, and a Provincial Instructor Diploma (PID) from Vancouver Community College (VCC).

Kwang Moo Yi is an assistant professor in the Department of Computer Science at the University of British Columbia (UBC), and a member of the Computer Vision Lab, CAIDA, and ICICS at UBC. Before, he was at the University of Victoria as an assistant professor. Prior to being a professor, he worked as a post-doctoral researcher at the Computer Vision Lab at the EPFL. He received his Ph.D. from Seoul National University. He also received his B.Sc. from the same University. He serves as area chair for top Computer Vision conferences (CVPR, ICCV, and ECCV), and AAAI. He was part of the organizing committee for CVPR 2023.

Klaske Van Heusden is an assistant professor at the School of Engineering (SoE) at the University of British Columbia. She holds an M.Sc. in control engineering from the Delft University of Technology and a Ph.D. from the EPFL in Lausanne, Switzerland. Prior to joining the UBC SoE, she was a postdoctoral fellow at the University of California, Santa Barbara and a research associate at the University of British Columbia. Her research interests include learning and data-driven control of safety-critical systems, vision-based control in manufacturing and control of highly uncertain systems with applications in agriculture and health care.

Guy A. Dumont, Professor and Distinguished University Scholar in the Department of Electrical and Computer Engineering at the University of British Columbia, is an Associate Member of UBC's Department of Anesthesiology, Pharmacology and Therapeutics. A Principal Investigator at BC Children's Hospital Research Institute, he co-directs the Digital Health Innovation Laboratory (DHIL). His research spans adaptive control, patient monitoring, signal processing for physiological monitoring, mobile health, and brain monitoring. His accolades include the IEEE Control Systems Technology Award, three NSERC Synergy Awards, and the Brockhouse Canada Prize. He is a Fellow of the IEEE, IFAC, and the Royal Society of Canada.